

Instructor: Date:

Learner: Regional Branch:

Program: Center:

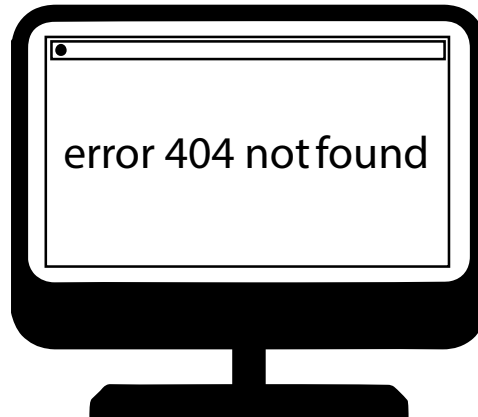
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Let's Start!

Data Validation as an Error Handling Tool



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain the concept of data validation and error handling.
- I can talk about data validation importance.
- I can describe different options of performing data validation.

Ask your partner:

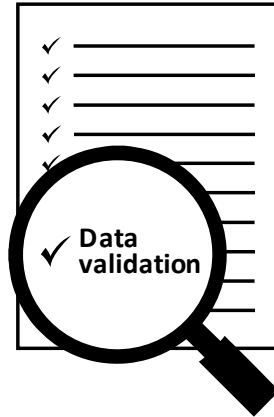
- Why do you think it is important to validate data when developing software?
- How do you think software designers can prevent users from entering invalid information in a program?





Let's Explore!

Alfred and Anna want to review some concepts before the exam



Instructions: Read the text below.

Alfred: Hey Anna! Are you studying for the exam?

Anna: Yes! I'm reviewing the concepts of data validation and error handling.

Alfred: So, what's error handling?

Anna: Um, Software engineers need to know how to handle errors to ensure quality software. Before they release a program, they have to test it in order to find as many errors as possible. After this testing process, they need to follow a four-stage process, which includes validation, detection, communication, and amelioration.

Alfred: You are right! The data validation process is the first line of defense when handling errors. So, tell me. How does data validation work?

Anna: Sure, data validation guarantees that a program operates according to specific parameters when the errors come from the front-end, that is, errors that the users make when they enter information, or when the system reads or inspects a disk file.

Alfred: Exactly. Software designers aim to guarantee the accuracy, consistency, and fitness of the software by means of data validation processes.

Anna: Now, it's my turn to ask. What kinds of data validation do you know?

Alfred: I know the simple but primitive data type validation, which attempts to verify the characters that the user enters are consistent with the ones in the database. I also have some knowledge about the simple range and constraint validation process that scans the range of the words the user inputs into the system.

Anna: What about the code and cross-reference validation method?

Alfred: Oh! Right. It attempts to verify whether the data is consistent with external rules or not; for example, if you type an email, it checks that you write it well and that it really exists.

Anna: Don't forget to mention the structured validation process, as it allows the combination of various basic data type validation steps with more complex processes.



Let's Practice!

Alfred explains some details about the data validation

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: aim to guarantee, need to follow, need to know .

1. Software engineers _____ how to handle errors to ensure a quality software.
2. Software developers _____ the accuracy, consistency, and fitness of the software by means of data validation processes.
3. Software developers _____ a four-stage process, which includes validation, detection, communication, and amelioration, in order to guarantee the proper functioning of the program.



Let's Practice!

Anna is improving her knowledge before the exam

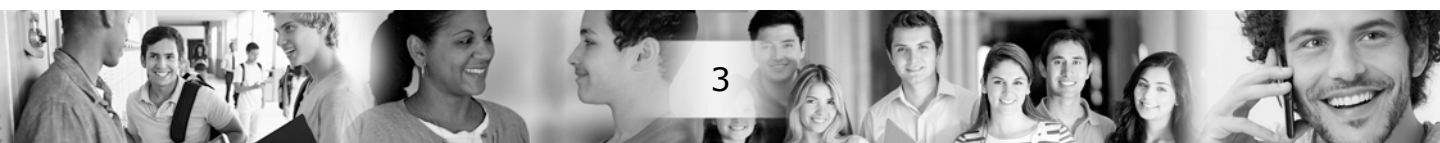
Instructions: Match Column A with Column B.

COLUMN A

1. Before releasing a program
2. Data validation is the process of guaranteeing
3. The simple range and constraint validation process

COLUMN B

- a. that a program operates on correct data.
- b. developers test the software in order to find as many errors as possible.
- c. scans the range of the words the user inputs into the system.





Extension Activity

Data validation process and rules setting

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for extra information about data validation processes and error handling to explain some basic rules.
- Write down at least 5 data validation rules people use to guarantee online forms to work.
- Share your findings with your instructor and classmates.

Self-Assessment

Yes No

1. I look for information about data validation and error handling.

☐ ☐

2. I make notes regarding data validation rules to ensure a form works.

☐ ☐

3. I share my findings with my classmates.

☐ ☐


Target Words

- **front-end** (adjective): the part of the program that the user can interact with
e.g. Data validation guarantees that a program operates according to specific parameters when the errors come from the front-end.
- **first line of defense** (expression): the first procedure or plan used to solve or detect an error
e.g. Data validation process is the first line of defense when handling errors.
- **guarantee** (verb): to ensure or to promise something
e.g. Software designers aim to guarantee the accuracy, consistency, and fitness of the software by means of data validation processes.
- **operate** (verb): to perform, to carry out, to work
e.g. Data validation guarantees that a program operates according to specific parameters.
- **disk file** (noun): a file stored in a magnetic disk
e.g. Users make errors when they enter information, or when the system reads or inspects a disk file.
- **attempt** (verb): to try to do something, usually something extremely difficult
e.g. It attempts to verify whether the data is consistent with external rules or not.



Learning Points

Action Verbs

We use **action verbs** when we want to express what someone or something does. In other words, just as its name suggests, **action verbs** refer to actions.

When we are working on data validation and error handling, we can use different **action verbs**. These verbs will help us describe the various steps or procedures we need to follow to complete these tasks.

Let's take a look at the following **action verbs** and their meaning.

Action verb	Definition	Example
Handle	To take care of something	We need to handle errors properly.
Release	To allow access, generally to a product, for the first time to the public	The company is going to release a new program.
Find	To locate or to discover the location of something	I need to find the error. This program is not working properly.
Correct	To fix or solve an error or a mistake	The team is correcting the program errors.
Test	To examine if something is working properly	We must test the code first.
Inspect	To check the current state of something	The team has to inspect the performance of this new software.
Analyze	To examine something, to go over something in detail	We can't analyze the software before the next stage.
Solve	To find a solution	Some programs are able to solve data errors without quitting.

Examples

- Users make errors when they enter information, or when the system reads or inspects a disk file.
- Before they release a program, they have to test it in order to find as many errors as possible.
- Data validation guarantees that a program operates according to specific parameters.
- The simple range and constraint validation process scans the range of the words the user inputs into the system.



Learning Points

Verbs + Infinitives

There are several verbs that demand the action that follows to be in its infinitive form.

Sometimes, the first verb can help determine if the action is a necessity, an obligation, a desire, etc.

For example:

I need to test the code. (Expressing necessity)

I want to buy the latest software. (Expressing desire)

The team **has to determine** the source of the error. (Expressing obligation)

Let's take a look at other verbs that oblige the following action to be in its infinitive form:

Verb	Example
Agree	The engineer agreed to release the software this summer.
Help	This new program helps you to determine where the error code is.
Decide	I decided to validate data by analyzing the code.
Learn	I learned to handle errors this way.
Plan	We are planning to divide the data validation task among all the team members.

Examples

- Software engineers need to know how to handle errors to ensure a quality software.
- Before releasing a piece of software, they have to test it in order to find as many errors as possible.
- Software designers aim to guarantee the accuracy, consistency, and fitness of the software by means of data validation processes.
- Don't forget to mention the structured validation process

What is a driver in computer science?

A **driver** is a software component that allows the operating system (software) and a device (hardware) to communicate with one another.





Learning Points

What is a driver in computer science?

A **driver** is a software component that allows the operating system (software) and a device (hardware) to communicate with one another.

Drivers enable communication between operating systems and devices, as the company that develops the operative system is not the same company that develops the devices; therefore, it needs a **driver** to allow the communication between them both. For example, if an application needs to read some data from a third party device, it communicates with the operating system, and the operating system communicates with the driver. As the driver is designed to communicate directly with the device, it gets the data from the device and returns it to the operating system; finally, the operating system transmits that information to the application.

However, not all drivers communicate directly with the device and in most cases, there are several drivers that communicate with other drivers until they reach the function driver, which is the one that communicates directly to the device.

On the other hand, we can also have software drivers.

These kind of drivers are not associated to any hardware at all, but to applications that commonly run on both user mode and kernel mode. That is, applications that, while working in user mode, have a unique process that prevents the whole system from collapsing in case the app crashes. On the other hand, all apps that run under the kernel mode share the same process; therefore, if one driver writes a wrong code, the whole operative system can be compromised. When an app is running under kernel mode, we call it software driver.

